Cointegration and Pairs Trading: AAPL, MSFT, and GOOG (2011–2025)"

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(a) Question of Interest

This study investigates whether a long-term equilibrium relationship exists among the stock prices of Apple Inc. (AAPL), Microsoft Corporation (MSFT), and Alphabet Inc. (GOOG) over the period 2011-07-01 to 2025-06-30, and whether this relationship can be exploited through a mean-reversion pairs trading strategy. The central research question is: ***Can cointegration among these three large-cap technology stocks be identified and utilized to generate consistent trading profits?***

This question is important for two reasons. First, cointegration-based trading provides a statistical arbitrage approach that is market-neutral, making it potentially resilient in varying market conditions. Second, technology stocks often exhibit correlated price movements due to sector-wide factors such as innovation cycles, macroeconomic conditions, and investor sentiment, yet temporary deviations from equilibrium may present profitable trading opportunities. Identifying and exploiting such deviations can contribute to portfolio diversification, risk management, and improved market efficiency by narrowing mispricings.

(b) Econometric methodology and why it is appropriate

I work with daily adjusted prices for AAPL, MSFT, and GOOG from 2011-07-01 to 2025-06-30. Because cointegration is defined for I(1) series, I first assess integration orders using two complementary unit-root tests: Augmented Dickey–Fuller (ADF) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS). The ADF fails to reject a unit root for each price series, while KPSS rejects level stationarity, supporting the I(1) assumption, hence a cointegration framework is appropriate.

Given three potentially I(1) series, I use Johansen's system approach to test for and estimate cointegration in log levels. I select the VAR lag length in levels by information criteria and adopt the AIC-preferred order (K=10) to balance bias and residual serial correlation in a daily setting. I estimate the Johansen trace test with a constant in the cointegrating space (ecdet="const"), which allows a non-zero mean in the long-run equilibrium without imposing deterministic trends in the short-run dynamics. The trace test indicates one cointegrating vector (r=1), so I proceed with a Vector Error-Correction Model (VECM) with r=1 to capture both long-run error-correction and short-run cross-effects.

For inference, I report Newey–West (HAC) robust t-tests on the stacked VECM coefficients to mitigate bias from potential heteroskedasticity and mild residual autocorrelation in daily data. I also run residual diagnostics on each VECM equation: (i) Ljung–Box tests at lag 20 (about one trading month) to check for left-over serial correlation, and (ii) ARCH LM tests at

lag 12 to screen for conditional heteroskedasticity typical of equity returns. These checks help validate that the chosen lag order is adequate for dynamics and that standard errors are robust to volatility clustering.

To link the econometrics to trading, I constructed a stationary spread from the estimated cointegrating vector. I normalize the vector on AAPL so that the spread is a static, self-financing portfolio: $s_t = logAAPL_t + b_{MSFT} logMSFT_t + b_{GOOG} logGOOG_t + c$. A simple mean-reversion rule generates positions: long the spread when s_t is below its historical mean by one standard deviation, short when above by one standard deviation, and neutral otherwise. Portfolio returns are computed from daily log-return changes using the hedge ratios implied by the cointegrating vector, with positions lagged one day to avoid look-ahead.

Performance is summarized with conventional statistics: average daily return, volatility, annualized Sharpe, maximum drawdown (from the log-equity curve), hit ratios (daily and per-trade), number of trades, and average holding period (computed from contiguous non-zero position runs). To reflect implementability, I apply explicit transaction costs: a per-side rate of 0.05% charged on every position change, including flips (e.g., long→short counts as two sides). I align the cost series to the return index, subtract costs from strategy returns, and recompute the full performance set on the cost-adjusted series. This delivers both raw and net (after-cost) metrics within a single coherent framework, ensuring that statistical evidence of cointegration translates into economically meaningful, implementation-aware results.

(c) Estimate your model and interpret your results

Unit root and stationarity tests

The Augmented Dickey–Fuller (ADF) tests fail to reject the null of a unit root for AAPL, MSFT, and GOOG (p-values 0.44, 0.87, and 0.36 respectively). The KPSS tests reject the null of level stationarity at the 1% level for all three series. Together, these results are consistent with each series being integrated of order one, I(1)I(1), satisfying the requirements for a Johansen cointegration analysis.

Johansen cointegration test

Using the AIC-selected VAR lag length of 10 in levels, the Johansen trace test with a constant in the cointegrating space reports one statistically significant cointegration relationship (r=1), with the r=0 trace statistic of 44.00 exceeding the 5% critical value

(34.91). This implies the existence of a long-run equilibrium linking the three log price series. The normalized cointegration vector (AAPL = 1) is:

$$Spread_t = \log(AAPL_t) - 1.3638\log(MSFT_t) + 0.5839\log(GOOG_t) - 0.1991$$

VECM estimation results

A rank-1 VECM with 9 lags of differenced terms (K=10 in levels) is estimated. The speed of adjustment coefficients on the error-correction term (ect1) are all negative and statistically significant at the 1% level:

- AAPL: −0.00665 (t ≈ −4.85, robust t ≈ −4.12)
- MSFT: −0.00501 (t ≈ −4.03, robust t ≈ −4.04)
- GOOG: −0.00390 (t ≈ −2.95, robust t ≈ −2.84)

These values indicate that when the spread deviates from equilibrium, each asset's return responds by partially correcting the disequilibrium the next day. MSFT and AAPL adjust fastest in absolute terms, suggesting they bear more of the burden of mean reversion.

Several short-run cross-lag effects are statistically significant. For example, MSFT.dl3 positively affects AAPL.d, and AAPL.dl7 positively affects MSFT.d, hinting at lead-lag spillovers. Robust Newey–West standard errors confirm that many of these effects remain significant after accounting for potential heteroskedasticity and mild autocorrelation.

Residual diagnostics

Ljung–Box tests at lag 20 for each equation show high p-values (> 0.83), suggesting no serious residual autocorrelation. However, ARCH LM tests at lag 12 strongly reject the null of no ARCH effects for all three equations ($p < 0.001$), consistent with volatility clustering in daily equity returns. This justifies the use of HAC-robust inference.

Trading simulation results (raw returns)

The cointegrating vector is used to form a mean-reverting spread with ±1σ trading bands. The strategy takes long (short) positions when the spread is sufficiently low (high), with positions lagged one day. Over the sample:

- Number of trades: 53
- Average daily return: 0.000529 (0.0529%)
- Daily return standard deviation: 0.01103

- Annualized Sharpe ratio : 0.761
- Maximum drawdown: 25.68%
- Hit ratio (daily): 16.2%
- Hit ratio (per trade): 100%
- Average holding period: ~19.85 days

The low daily hit ratio but perfect per-trade hit ratio indicates the strategy's profits come from sustained mean-reversion moves rather than frequent daily wins.

Cost-adjusted performance

Applying a 0.05% per-side transaction cost to every position change (including flips), there are 106 position change events over the sample. Total transaction costs amount to ~0.053 in cumulative return units. After costs:

- Average daily return: 0.000514
- Annualized Sharpe ratio: 0.739

While transaction costs modestly reduce performance, the net Sharpe remains close to the raw value, indicating that the trading rule is relatively robust to realistic costs.

(d) How the results answer the question, economic implications, limitations, and recommendations

The objective was to determine whether AAPL, MSFT, and GOOG share a stable long-run relationship and, if so, whether this relationship can be profitably exploited through a mean-reversion trading strategy.

Existence of a stable long-run relation.
The Johansen trace test (AIC-selected lag length K=10K=10) identifies one cointegrating vector (r=1), indicating a persistent equilibrium among the three stocks. The estimated relation, $\log(AAPL_t) - 1.3638\log(MSFT_t) + 0.5839\log(GOOG_t) - 0.1991$, is economically plausible given the firms' similar sectoral exposures and market leadership. In the VECM, all error-correction terms (ECTs) are negative and statistically significant under both conventional and Newey–West robust inference, confirming that deviations from equilibrium tend to revert over time.

Practical exploitability via mean reversion.
A ±1σ threshold strategy on the cointegration spread (with a one-day lag to avoid lookahead bias) produced:

- **Trades:** 53 over the sample, with an average holding period of ~19.85 days.
- **Returns:** Avg daily = 0.000529; annualised Sharpe ≈ 0.761; max drawdown ≈ 25.68%.
- **Hit ratios:** Daily = 16.2%, but trade-level = 100% (multi-day reversion captures gains).

After applying realistic transaction costs (0.05% per side on every position change), performance remains positive (avg daily = 0.000514; Sharpe ≈ 0.739) despite 106 position change events and total costs ≈ 0.053 in cumulative return units. This resilience suggests the signal is economically meaningful and not an artefact of cost-free simulation.

Economic implications for investors.

- The persistence of the cointegration relationship implies that these large-cap tech stocks are influenced by common macro and sector-specific fundamentals, making relative mispricings predictable in the short to medium term.
- The strategy's average holding period (~3 weeks) and reliance on mean reversion suggest it is best suited for **medium-horizon trading** rather than ultra-short-term scalping or long-term investing.
- Long-term investors could use the spread as a relative-value indicator accumulating underpriced stocks and trimming overpriced ones, while active traders could employ the full trading rule to capture temporary dislocations.

While the findings are promising, they also highlight several limitations that could affect practical implementation:

1. **Volatility clustering and ARCH effects:** Significant ARCH effects in residuals suggest that risk is time-varying; ignoring this may underestimate tail risks.
2. **Structural stability:** The model assumes the cointegration relationship is constant; structural breaks from regime changes, major tech sector disruptions, or macro shocks (e.g., COVID-19) could weaken predictive power.

3. **Sample and overfitting risk:** All results are in-sample; performance may decline out-of-sample if thresholds or model parameters are too closely fitted to historical data.
4. **Execution frictions:** While conservative transaction costs are included, market impact, slippage, and liquidity constraints especially during high-volatility periods could reduce realised performance.

Given these limitations, there are several natural extensions to strengthen the analysis:

- **Time-varying cointegration:** Use rolling or recursive estimation to detect and adapt to shifts in the long-run relationship.
- **Alternative trading thresholds:** Explore asymmetric or volatility-adjusted bands (e.g., 1.5σ, 2σ) to optimise trade frequency and risk-adjusted returns.
- **Portfolio expansion:** Test inclusion of additional correlated assets (e.g., NVDA, META) to diversify risk and improve signal robustness.
- **Risk-adjusted position sizing:** Incorporate dynamic sizing based on volatility forecasts (e.g., GARCH) to mitigate drawdowns during turbulent periods.
- **Out-of-sample and live testing:** Validate the strategy on a hold-out period or in paper-trading to assess robustness before deployment.

Conclusion.
The results confirm both statistical cointegration and economically meaningful mean reversion among AAPL, MSFT, and GOOG, supporting profitable medium-term relative-value trading. While promising, the approach must account for structural shifts, execution realities, and volatility dynamics to ensure robustness in live market conditions.

Appendix

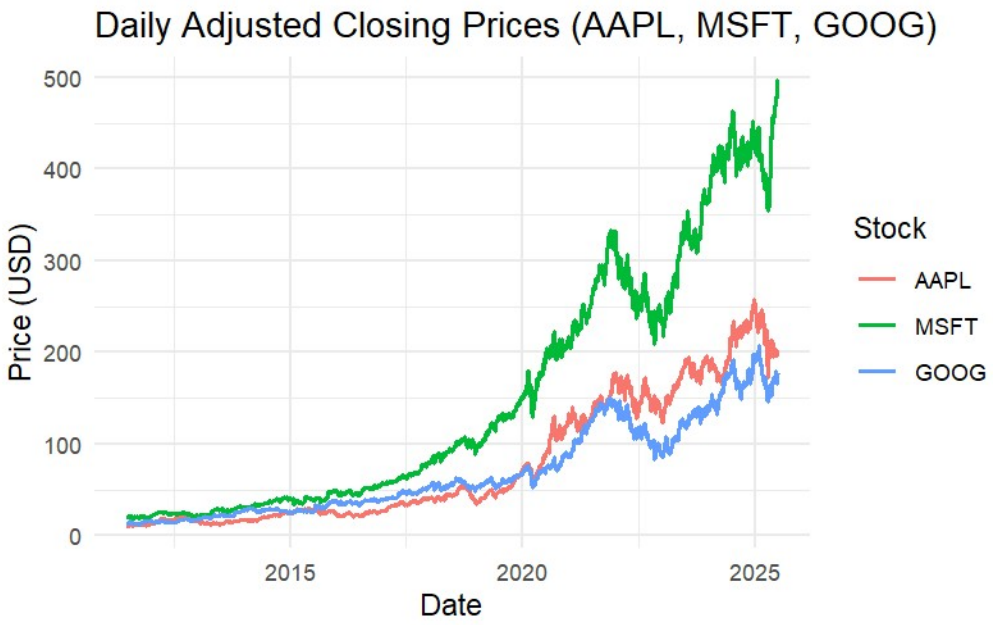

Figure 1: Daily Adjusted Closing Prices (AAPL, MSFT, GOOG)

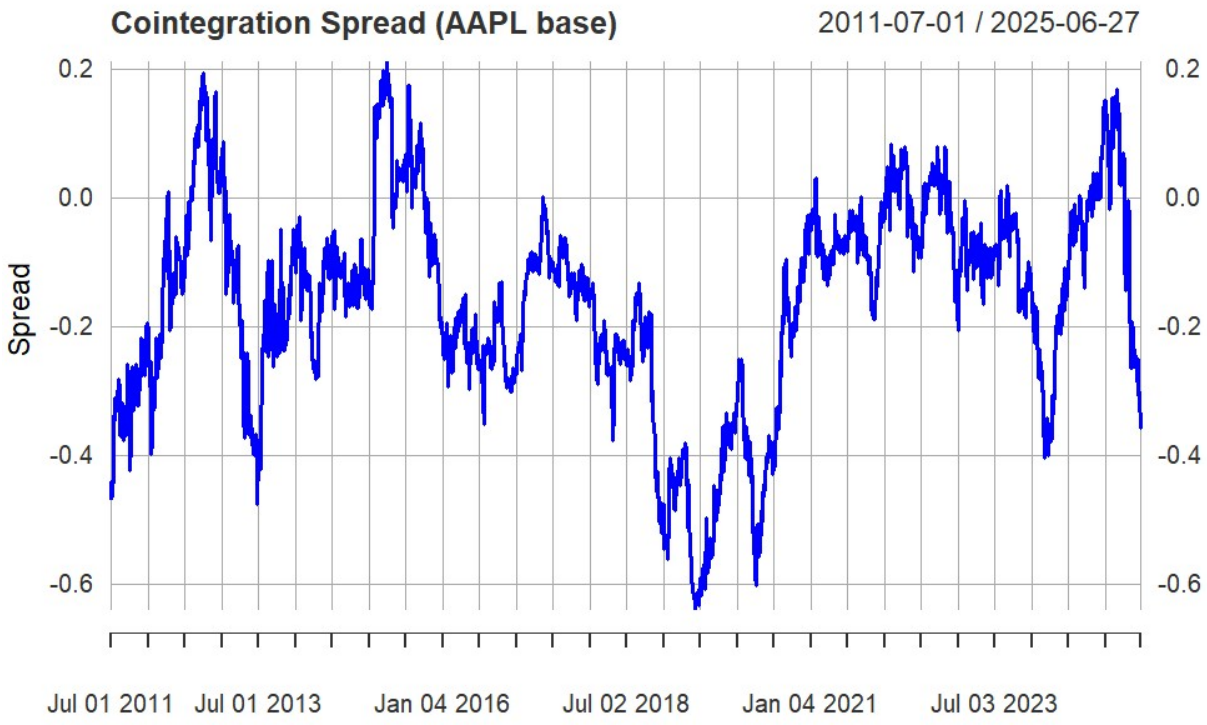

Figure 2: Cointegration Spread (AAPL Base) with ±1σ Bands

Figure 3: Cumulative Log Returns from Mean-Reversion Trading Strategy